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AdaKoop: Efficient Modeling of Nonlinear Dynamics from Nonstationary Data Streams with Koopman Operator Regression

Naoki Chihara Ren Fujiwara Yasuko Matsubara Yasushi Sakurai

SANKEN, The University of Osaka

Code Paper Website LinkedIn

Motivation

- **Real-time data analysis** requires the ability to accurately and adaptively address nonlinear dynamics in a non-stationary data stream while preserving computational efficiency.
- To make nonlinear dynamics computationally tractable, we utilize **Koopman operator theory**, representing them linearly in an infinite-dimensional space.
- We present **AdaKoop**, an efficient streaming algorithm for modeling nonlinear dynamics over nonstationary data streams.

Model

We formulate a probabilistic state space model that treats both raw observations and RKHS features as emissions from a shared latent linear dynamical system.

Our Model

MODEL 1. *Let $\mathbf{z}_t \in \mathbb{R}^r$ be the latent vector at time t, and $\mathbf{y}_t \in \mathbb{R}^{d+m}$ the augmented observation vectors at time t. The following equation governs the single nonlinear dynamic pattern.*

$$\begin{aligned} \mathbf{z}_{t+1} &= \mathbf{A}\mathbf{z}_t + \boldsymbol{\xi}_t, \quad \boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \\ \mathbf{y}_t &= \mathbf{H}\mathbf{z}_t + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) \end{aligned} \tag{2}$$

with initial conditions $\mathbf{z}_1 \sim \mathcal{N}(\boldsymbol{\mu}_0, \mathbf{P}_0)$

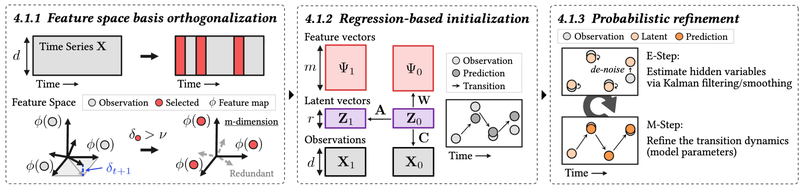
- $\mathbf{y}_t = \begin{bmatrix} \mathbf{x}_t \\ \psi(\mathbf{x}_t) \end{bmatrix} \in \mathbb{R}^{d+m}$: augmented observation vector
- $\psi : \mathbb{R}^d \to \mathbb{R}^m$: feature map associated with an RKHS

Model Parameter for Single Dynamic Pattern

DEFINITION 1 (DUAL-VIEW KERNELIZED SYSTEM: DKS). *Let $\boldsymbol{\theta}$ be parameters of a dual-view kernelized system for a single nonlinear dynamic pattern, i.e., $\boldsymbol{\theta} = \{\mathbf{A}, \mathbf{Q}, \mathbf{C}, \mathbf{W}, \mathbf{R}_x, \mathbf{R}_\psi, \boldsymbol{\mu}_0, \mathbf{P}_0\}$.*

Optimization Algorithm

Our static optimization consists of the following elements:



1. Feature space basis orthonalization

We determine whether each data point is selected.

$$\delta_{t+1} = \left\| \phi(\mathbf{x}_{t+1}) - \Pi_{\mathcal{D}_t} \phi(\mathbf{x}_{t+1}) \right\|^2_{\mathcal{H}},$$

The selected data is used for feature vector.

$$\psi(\mathbf{x}) = [k(\mathbf{x}_i, \mathbf{x})]^\top_{i \in \mathcal{I}_T} \in \mathbb{R}^m$$

2. Regression-based initialization

We estimate the corase parameter $\{\tilde{\mathbf{A}}, \tilde{\mathbf{C}}, \tilde{\mathbf{W}}\}$ based on the Koopman operator regression.

3. Probabilistic refinement

We use the EM algorithm, initialized with $\{\tilde{\mathbf{A}}, \tilde{\mathbf{C}}, \tilde{\mathbf{W}}\}$. Thanks to linearity, we can avoid computationally heavy inference.

Streaming Algorithm

1. Statistical hypothesis testing-based model selection

We detect the abrupt changes based on the CUSUM test.

$$g_{t_c} = \max\left\{0,\ g_{t_c-1} + \left(e^2_{t_c} - \chi^2_d(1-\alpha)\right)\right\}.$$

The value g_{t_c} exceeding a switching limit indicates that we should search for a better model.

2. Adaptive parameter update

Given a new data $\mathbf{x}_{t_c}$, we decide whether to be selected based on δ_{t_c}. Then, we update the model parameters incrementally using a one-step online EM with a forgetting factor to handle the gradual changes.

Experimental Results

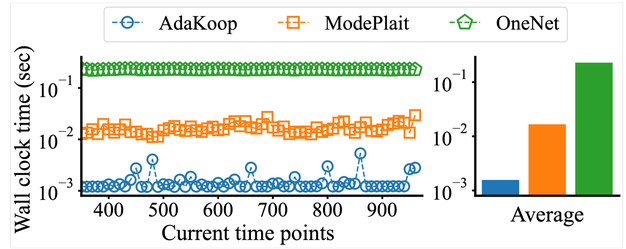
We used the dysts benchmark [Gilpin 21], which offers diverse and practical dynamical systems.

Q1. Accuracy. How accurately does it forecast future values?

Forecasting results averaged across 71 datasets

Metrics	*Mean Squared Error (MSE)*			*Mean Absolute Error (MAE)*		
Steps	20	25	30	20	25	30
sKAF (2023)	34.0 ± 112	59.6 ± 285	81.4 ± 270	1.20 ± 1.96	1.41 ± 2.35	1.72 ± 2.61
Koopa (2023)	0.397 ± 0.417	0.377 ± 0.331	0.361 ± 0.338	0.430 ± 0.165	0.427 ± 0.154	0.423 ± 0.144
OneNet (2023)	0.318 ± 0.148	0.321 ± 0.146	0.322 ± 0.145	0.441 ± 0.123	0.443 ± 0.122	0.444 ± 0.121
PAttn (2024)	0.311 ± 0.206	0.326 ± 0.206	0.330 ± 0.196	0.407 ± 0.139	0.417 ± 0.138	0.421 ± 0.134
WPMixer (2025)	0.561 ± 1.95	0.509 ± 1.56	0.436 ± 0.916	0.411 ± 0.233	0.413 ± 0.214	0.413 ± 0.194
ModePlait (2025)	0.279 ± 0.0918	0.300 ± 0.104	0.325 ± 0.115	0.407 ± 0.0857	0.423 ± 0.0914	0.441 ± 0.0964
AdaKoop (Ours)	**0.0763 ± 0.0539**	**0.0999 ± 0.0659**	**0.120 ± 0.0747**	**0.183 ± 0.0769**	**0.214 ± 0.0857**	**0.240 ± 0.0888**

Q2. Scalability. How does it scale in computational time?



AdaKoop consistently outperformed its competitors

Q3. Nonlinear capability. How well does it estimate and leverage complex nonlinear dynamics in a streaming fashion?

Metrics		MSE	MAE
RBF	20	0.0763 ± 0.0539	0.183 ± 0.0769
	25	0.0999 ± 0.0659	0.214 ± 0.0857
	30	0.120 ± 0.0747	0.240 ± 0.0888
Polynomial	20	0.115 ± 0.0922	0.221 ± 0.0953
	25	0.144 ± 0.108	0.254 ± 0.102
	30	0.164 ± 0.114	0.278 ± 0.103
Sigmoid	20	0.218 ± 0.0868	0.365 ± 0.0837
	25	0.239 ± 0.0967	0.384 ± 0.0889
	30	0.253 ± 0.101	0.398 ± 0.0920
Linear	20	0.218 ± 0.0870	0.366 ± 0.0846
	25	0.239 ± 0.0963	0.386 ± 0.0893
	30	0.253 ± 0.0999	0.400 ± 0.0917

Ablated variants in our settings

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naoki88@sanken.osaka-u.ac.jp